Samuel Crabbendam

crabbendam.net/sam | US Citizen | 407-973-5067 | linkedin.com/in/samuel-crabbendam-93ab33349/ | crabbendams@gmail.com

Experience

Electric Fixed Wing UAV Design & Build | *Independent Project*
Gainesville, FL | July **2026** – Present

- Designing & assembling a 5-ft electric PX4 GPS-guided UAV, integrating wing, avionics, radio & propulsion systems.
- Wiring & soldering onboard electronics: 6S, 1390 W EMAX 2807 1300KV motor, 60 AMP ESC, receiver & servos.
- Validated the NACA 2412 wing in Flow5, confirming lift exceeds 1.5 kg weight at 13 m/s design airspeed.
- Cut airframe mass 40% from 2.5 to 1.5 kg by redesigning the fuselage from PLA to foamboard via a material trade study.
- Designed a bolted, adhesive-free wing mount with 3D printed spar clamps, offsetting spar holes to set wing dihedral.
- Planned ROS2 computer integration with PX4 for offboard test maneuvers & flight data logging to improve model iterations.

Payload Mechanics Responsible Engineer | *Swamp Launch Rocketry, IREC Team*
Gainesville, FL | August **2024 –** Present

- Led a 6-person subteam integrating a rotating sensor module for the 2027 Intercollegiate Rocket Engineering Competition (IREC) payload from concept to detail in Fusion360 CAD, holding ± 0.01 in. sealing tolerances; ICDs approved at team PDR.
- Updated drawings & BOMs, documented configurations & revisions, applying DFM to take prototype 1 to manufacturing.
- Engineered thermoelectric thermal management system to hold bio payloads within ±5 °C of 30 °C under 6 g launch loads.
- Selected PETG for the payload structure via a trade study of heat deflection (75°C), glass transition & cost.

Experimental Aerodynamics Test Engineer | *Wind Tunnel Undergraduate Research Group, University of Florida*
Gainesville, FL | August **2026** – Present

- Characterizing flow & calibrating a Velmex 3-axis pitot-static sensor assembly at up to 20 m/s for model flight testing.
- Root-caused & resolved positioner fit interference with a CNC milled test section insert, providing ± 180° yaw clearance.
- Fabricated a Unistrut frame (10 members cut to 20 ± 0.1 in.) for a 4-axis model positioner under a 3'5" wind tunnel test section.

Project Engineering Intern | *Garney Construction*
Tampa, FL | May **2026** – August **2026**

- Verified GD&T on assembly drawings for 11 concrete assemblies, confirming ± 2 in. design-to-production tolerances.
- Implemented an Excel data tool to track procurement, reducing production delays by 20% weekly across field & vendor teams.
- Performed QC inspections on four structural assemblies per week, locating deviations with field crews.
- Authored 4 technical reports for life cycle verification & processed 15 change orders & submittals across vendor teams.

Education

University of Florida, Wertheim College of Engineering **Expected May 2028**

Bachelor of Science in Aerospace Engineering GPA: 3.6

Coursework: *Finite Element Analysis (Spring 2027), Design & Manufacturing, Aerodynamics, Mechanics of Materials*

Certifications

- Part 61 Private Pilot License, *Orlando Flight School, **55 hours*** Orlando, FL | 2026 - Present
- CSWA Certified, *SolidWorks* Gainesville, FL | 2025 - Present
- Part 107 Remote Pilot License, *West Orange High School* Winter Garden, FL | 2023 - Present

Skills

- **3D CAD & Drafting:** SolidWorks, Fusion360, Assembly Drawings, ASME Y14.5 GD&T, Tolerances
- **UAV Systems:** UAV Assembly, PX4, Sensor & Communication Integration, Bench Testing, ROS2 (planned)
- **Hardware:** Rapid Prototyping, 3D Printing, Electric Propulsion, Soldering & Wiring, Test Rigs, Root Cause Analysis
- **Analysis:** FEA (ANSYS), MATLAB, Carbon Fiber Spar Sizing, Structural Loads, Trade Studies, Flow5, Excel